
A/B Testing Revisited

Classical vs Bayesian Approaches
to Conversion Lift in Udacity Data

Insights by-

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Introduction :

A/B testing is a statistical experiment used to compare two versions of a product or webpage to determine which performs better. Udacity released a public dataset containing user interactions from an online experiment designed to measure whether a new webpage increased student conversions. In this project, the business objective was to determine whether the new layout improves the conversion rate, i.e., the probability that a user signs up.

To answer this, we conduct both **classical (frequentist)** and **Bayesian** analyses. The classical approach evaluates whether the observed difference in conversion rates is statistically significant using hypothesis testing and confidence intervals. The Bayesian approach provides a probability-based interpretation by estimating posterior conversion rates, the probability that the treatment outperforms the control, the expected loss of deploying the treatment, and credible intervals for the lift.

This combined investigation allows us to compare conclusions from both statistical frameworks, assess the reliability of the experiment, and understand whether Udacity collected sufficient data to make an informed decision. The results offer a comprehensive view of the treatment's impact and the risks associated with adopting the new webpage design.

Abstract

This project explores two major statistical approaches—**Classical (Frequentist) A/B testing and Bayesian A/B testing**—to evaluate potential improvements in user conversions for Udacity’s landing page experiment. The dataset contains user-level information on page assignment and conversion behaviour, allowing us to compare the performance of the existing page with the redesigned version. **The classical framework uses hypothesis testing and confidence intervals to assess whether any observed difference is statistically significant.** In contrast, the **Bayesian framework focuses on estimating the probability that one version outperforms the other, quantifying uncertainty more directly, and providing decision-oriented measures such as expected lift and risk.**

Through these two approaches, **the project aims to understand not only how the two versions compare, but also how the choice of statistical method influences interpretation, reliability, and business decision-making.**

Beyond the experiment itself, the study highlights the broader value of Bayesian methods in real-world settings due to their flexibility, intuitive probability-based insights, and ability to incorporate prior information. Ultimately, this project demonstrates how combining classical and Bayesian viewpoints offers a more complete and practical foundation for evaluating product changes in data-driven environments.

Objective:

1. To compare the conversion rates of control and treatment groups.
2. To test whether the new webpage significantly improves conversions.
3. To perform both:
 - **Classical (frequentist) hypothesis testing**
 - **Bayesian A/B testing**and figure out which one is better for analysis.
4. To check **sample balance**, **effect heterogeneity**, and **statistical power**.
5. To provide actionable business recommendations.

About the dataset

The **Udacity A/B Test dataset** contains user-level data from an online experiment designed to test whether a **new course landing page** leads to higher student conversion compared to the old page. The experiment randomly assigned users into two groups:

VARIABLES	MEANING
User id	Unique user identifier
group	A/B assignment (control or treatment)
landing_page	Which page the user actually saw (old/new)
converted	Whether the user enrolled (1 = converted, 0 = did not convert)

- **Control Group** → Saw the *old* landing page
- **Treatment Group** → Saw the *new* landing page

N.B: You can view the dataset [ab_data.csv](#)

Framework and Statistical Foundations of A/B Testing

Frequentist (Classical) A/B Testing – Theory Overview

Frequentist or classical A/B testing is one of the most widely used statistical frameworks for comparing two versions of a product, webpage, or intervention to determine which performs better. At its core, it **relies on probability theory, sampling distributions, and hypothesis testing to make decisions based on observed data**. The approach assumes that the underlying parameters—such as conversion rates—are fixed but unknown, and the randomness arises solely from the data collected through repeated sampling.

In an A/B test, users are randomly divided into two independent groups: a control group (A) that receives the existing version, and a treatment group (B) that receives the new variant. Each user outcome, such as clicking a button or completing a purchase, is typically **modelled as a Bernoulli trial**. Aggregating these outcomes across users gives rise to binomial distributions for both groups. The primary question is whether the difference in conversion rates between the two groups is large enough to rule out the possibility that it occurred merely by chance.

The classical **method formalizes this through hypothesis testing**. The null hypothesis (H_0) generally states that there is no difference between the two conversion rates, while the alternative hypothesis (H_1) suggests that a meaningful difference exists. A test statistic—commonly the difference in sample proportions—is computed and standardized. Under the null hypothesis, this statistic follows a known distribution, allowing the calculation of a p-value, which quantifies the probability of observing a difference as extreme as the one seen, assuming H_0 is true.

A small p-value (typically less than 0.05) provides evidence against the null hypothesis, suggesting that the treatment may indeed differ in performance. Alongside p-values, frequentist analysis uses confidence intervals, which provide a range of plausible values for the difference in conversion rates. If this interval excludes zero, it is interpreted as further evidence of a meaningful effect.

Despite its widespread use, the classical approach has limitations. **It does not provide the probability that one version is better than the other**; rather, it addresses the probability of observing the data under a specific assumption. It **also relies heavily on large sample sizes to achieve adequate statistical power. Underpowered experiments, common in A/B testing, often lead to inconclusive results or false negatives.** Moreover, frequentist testing traditionally requires the sample size to be fixed in advance; peeking at results midway can inflate error rates unless corrected.

Nevertheless, Frequentist A/B testing remains a foundational and practical framework due to its simplicity, interpretability, and strong theoretical grounding. When the assumptions are met and the experiment is properly designed, it offers a reliable and objective way to assess whether a proposed change leads to a statistically significant improvement.

Bayesian Testing – Theory Overview:

The main objective of a business is to maximize revenue. Conducting controlled experiments is an effective way to do so.

Let us take an example of an online e-commerce business with an objective of increasing its transactions. It currently has a green “purchase” button on its webpage and wants to experiment by changing the colour to blue and see if this change has an effect on the number of transactions.

We would call the page with the green button variant A (the control) and the one with blue variant B (the treatment). In this paper, we focus on conversion-base experiments. Therefore, every experiment has trials and successes. Each visitor to the webpage counts to a trial and each purchase counts to a success (we also refer to a success as a conversion). The measure of interests are the uplift in conversion rate, and revenue-based metrics like average order value and customer lifetime value.

In the execution of the experiment, each visitor is randomly assigned to variant A or variant B. Given a valid design and a correct execution, the only difference between the two variants is the colour of the button; all other factors such as seasonality or moves by the competition occur in both control and treatment. This means, that the difference in the measures of interests are either due to the change of the button colour or by random chance . To determine the “relevance” of the difference, statistical methods are used—either a statistical test in terms of the Frequentist inference or, as we motivate here, a model driven approach in terms of Bayesian inference.

In detail, we present a Bayesian approach to answer the quintessential questions that typically arise in the mind of a business owner:

- 1) What is the probability that variant B is better than variant A?
- 2) How much uplift in conversions will variant B bring?
- 3) What is the risk of switching to variant B from variant A?

The Bayes’ theorem is the fundamental model describing such updating:

$$P(\lambda|X) = \frac{P(\lambda) P(X|\lambda)}{P(X)}$$

It describes the dependency of the posterior distribution $P(\lambda|X)$ of a parameter λ after seeing data X with the prior distribution $P(\lambda)$ of the parameter λ , the likelihood $P(X|\lambda)$ of the data X given the parameter λ , and the marginal likelihood $P(X)$ of the data X . In many cases, also in this paper, we can ignore the marginal likelihood and work with

$$P(\lambda|X) \propto P(\lambda)P(X|\lambda)$$

With this, the primary task in developing Bayesian models for specific applications is to define proper models for the prior and the likelihood. A very important concept in Bayesian statistics is conjugacy, meaning that the posterior distribution $P(\lambda|X)$ is in the same probability distribution family as the prior distribution $P(\lambda)$. The prior is then called a conjugate prior. This allows us to write the posterior in a closed-form expression which is mathematically as well as computationally convenient.

The binomial distribution $\text{Bin}(n,p)$ is a discrete distribution defined on the interval $\{0, \dots, n\}$. It describes the number of successes in $n \in \mathbb{N}$ trials with success probability $p \in [0,1]$ in each trial. The Beta distribution $\text{Beta}(a,b)$ is a continuous distribution defined on the interval $[0, 1]$. The parameters a and b define the shape of the distribution, e.g., $a = b = 1$ gives the standard uniform distribution. The Beta distribution is the conjugate prior for the binomial distribution.

The experiment consists of variants i with $i = 1, \dots, V$ where V is the total number of variants. Each variant displays only one option to the visitor. For a given variant i , N_i is the total number of visitors that saw variant i (trials), and C_i the number of conversions (successes). Success is defined by the occurrence of the event of interest, for example, the visitor clicks a button click or purchases a product. A success also has a value v_i such as revenue or customer lifetime value associated and can be different for every variant. In a frequentist

approach, the conversion rate λ_i is the point estimate C_i/N_i , the total revenue earned is $N_i \cdot \lambda_i \cdot v_i$, and the revenue earned per visitor per variant is $\lambda_i \cdot v_i$.

Executing the experiment results in collected data for each variant, $X_i = \{x_{i1}, x_{i2}, \dots, x_{iN_i}\}$ with $X_{ij} \in \{0, 1\}$ $j = 1, \dots, N_i$ and 0 indicates a non-conversion and 1 a conversion. Since the conversions are Boolean-valued, the sequence of conversions X_i follows the binomial distribution and the generative process of the data is:

$$P(X_i | \lambda_i) = \text{Bin}(N_i, \lambda_i)$$

The prior distribution about the conversion rate is modeled as

$$P(\lambda_i) = \text{Beta}(a_i, b_i)$$

To compute the posterior, we leverage the fact that the conjugate prior for a distribution belonging to binomial family is in the beta family, and therefore the posterior distribution is the same as the prior distribution with updated parameter

$$P(\lambda_i | X_i) = \text{Beta}(a_i + C_i; b_i + (N_i - C_i))$$

The posterior $P(\lambda_i | X_i)$ gives the probability distribution of all possible values of λ_i given the evidence X_i . We can draw n random samples from the posterior to obtain a set

$Y_i = [y_{i1}, \dots, y_{in}]$ of possible values the conversion rate λ_i can take. The possible values for the revenue (or any other value $v_i > 0$) per visitor is defined as $Z_i = Y_i \cdot v_i$

There are cases where businesses incur loss for each no conversion. One example is the pay-per-click model followed in online advertising industry where businesses must pay the ad platform for each click generated regardless of conversion.

The above model can be adjusted to penalize the loss l_i of each non-conversion as

$$Z_i = Y_i \cdot (v_i - l_i) + (1 - Y_i) \cdot (-l_i)$$

Theory Used for The Analysis:

Bernoulli & Binomial Model:

Each user either converts (1) or not (0).

For n users:

$X \sim \text{Binomial}(n, p)$ [p is the conversion probability]

Bayesian Model:

We place a Beta(1,1) prior (uniform)

Posterior of each group:

$\text{Posterior}_{\text{control}} = \text{Beta}(1+x_c, 1+n_c-x_c)$

$\text{Posterior}_{\text{treatment}} = \text{Beta}(1+x_t, 1+n_t-x_t)$

Understanding the Differences Between Bayesian and Frequentist Statistics:

Highlights:

- There are 2 main approaches to statistical inference, frequentist and Bayesian, differing in their interpretation of uncertainty.
- The frequentist approach deals with long-run probabilities (i.e., how probable is this data set given the null hypothesis), whereas the Bayesian approach deals with the probability of a hypothesis given a particular data set.
- Bayesian analysis incorporates prior information into the analysis, whereas a frequentist analysis is purely driven by the data.
- The Bayesian approach can calculate the probability that a particular hypothesis is true, whereas the frequentist approach calculates the probability of obtaining another data set at least as extreme as the one collected (giving the P value).

Let us **understand this topic with an example :**

Flip a coin.

Before you look at the result, pause and ask:

What's the **probability** the coin landed on heads?

Depending on your answer, you're either a Frequentist or a Bayesian—statistically speaking, at least.

According to a **Frequentist statistical approach**, there's a single correct answer. If the coin is heads, the probability that the coin landed on heads is 100%. If it's tails, the probability is 0%.

With **Bayesian statistics**, probabilities are more subjective. In the coin toss example, a Bayesian approach states that the probability of it landing on heads or tails boils down to your beliefs. You might believe there's a 50% chance for each outcome, or your confidence in the coin's fairness could also influence your answer. According to this statistical approach, once you've seen the result, you'd update your beliefs based on that new information.

The main difference between the two methodologies is how they handle uncertainty. Frequentists rely on long-term frequencies and assume that probabilities are objective and fixed. Bayesians embrace subjectivity and the idea that probabilities change based on new information.

Key Takeaways:

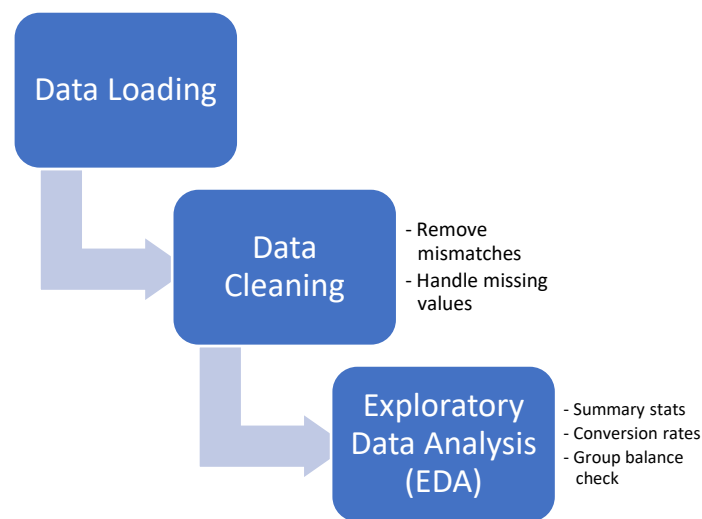
Topic	Frequentist	Bayesian
Hypothesis Testing	Set null and alternative hypotheses and use statistical tests to assess evidence against the null.	Consider prior beliefs when forming hypotheses.
Probability Interpretation	Frame probability in terms of objective, long-term frequencies.	Interpret probabilities subjectively and update them as new data is collected.
Sampling	Emphasize random sampling and often require fixed sample sizes.	Can adapt well to varying sample sizes since Bayesians update their beliefs as more data comes in.

Methodology:

A/B TESTING METHODOLOGY FLOWCHART

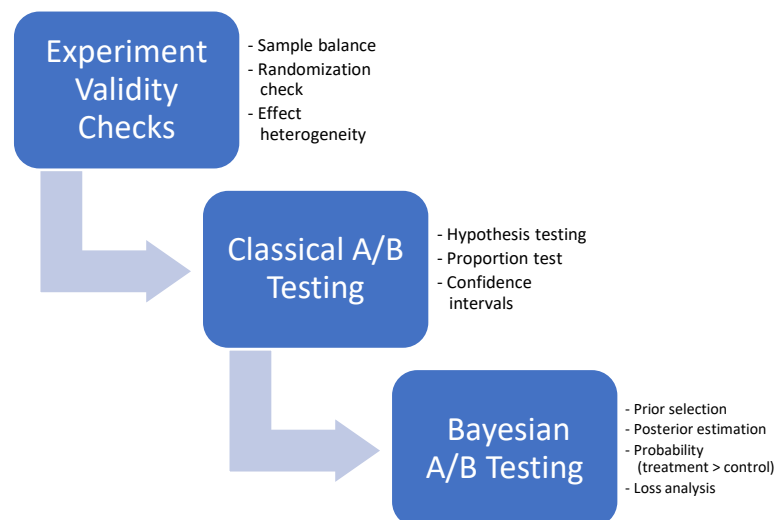
A. RAW DATA ANALYSIS

Before testing the hypothesis, the raw dataset is examined to confirm its quality, remove inconsistencies, and prepare a reliable foundation for the A/B analysis.



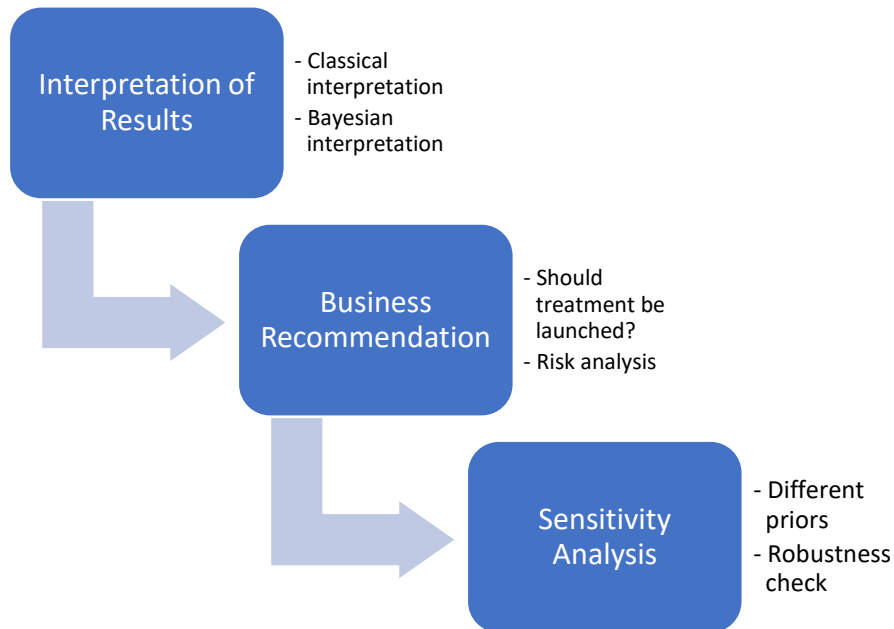
B. TESTING

In this stage, we conduct statistical tests—frequentist and Bayesian—to evaluate the significance, probability, and reliability of the observed differences between the two groups.



C. SENSITIVITY ANALYSIS AND CONCLUSION

This section summarizes the overall findings from the analysis and concludes whether the treatment provides any meaningful improvement over the control.



This methodology ensures that the dataset is cleaned, validated, and analysed through both classical and Bayesian frameworks. Together, these steps provide a robust foundation for drawing reliable conclusions about the experiment.

R codes

Exploratory Data Analysis (EDA) is the first step in understanding the dataset. It helps to summarize the main characteristics of the data, check for missing values, and identify patterns or anomalies. For the Udacity A/B Test dataset, EDA involves examining the structure of the data, summarizing numeric and categorical variables, visualizing distributions, and exploring relationships between features. This process provides insights that guide further analysis and modelling.

```
# EDA
df <- read.csv("C:/Users/CINI CHETANA/OneDrive/Desktop/Project with Supro/ab_data.csv")

head(df)
summary(df)

# missing
colSums(is.na(df))

# conversion by landing page
tapply(df$converted, df$landing_page, mean)

# mismatch check
df$match <- ifelse(df$group=="control" & df$landing_page=="new_page", 1,
                  ifelse(df$group=="treatment" & df$landing_page=="old_page", 1, 0))

sum(df$match)

# removeing mismatches
df_clean <- df[df$match==0, ]
```

The following R code performs a classical A/B test using a two-sample proportion test. It calculates conversion rates for both groups, runs the hypothesis test, and reports the confidence interval for the difference in proportions.

```
## Classical A/B Testing
df <- read.csv("C:/Users/CINI CHETANA/OneDrive/Desktop/Project with Supro/ab_data.csv")
df
# conversions and sample sizes
```

```

c_conv <- sum(df$converted[df$group=="control"])
t_conv <- sum(df$converted[df$group=="treatment"])

n_c <- sum(df$group=="control")
n_t <- sum(df$group=="treatment")

# conversion rates
p_c <- c_conv / n_c
p_t <- t_conv / n_t
p_c
p_t

# two-sample proportion test
prop.test(c(c_conv, t_conv),
          c(n_c, n_t),
          alternative="two.sided",
          correct=FALSE)

# confidence interval
prop.test(c(c_conv, t_conv),
          c(n_c, n_t),
          correct=FALSE)$conf.int
# bar plot of conversion rates
barplot(c(p_c, p_t),
        names.arg=c("Control", "Treatment"),
        main="Conversion Rates",
        ylab="Rate",
        col=c("skyblue", "salmon"))

# difference distribution (bootstrap)
set.seed(123)
boot_diff <- replicate(5000, {
  c_samp <- sample(df$converted[df$group=="control"], n_c, replace=TRUE)
  t_samp <- sample(df$converted[df$group=="treatment"], n_t, replace=TRUE)
  mean(t_samp) - mean(c_samp)
})

plot(density(boot_diff),
     main="Bootstrap Distribution of Difference (Treatment - Control)",
     xlab="Difference in Conversion Rate")
abline(v=0, col="red", lwd=2)

```

The following R code checks whether the A/B experiment was properly executed. It examines sample balance between groups, looks for effect heterogeneity across user segments, and computes the statistical power based on the observed conversion rates. This helps verify the reliability of the experiment before interpreting the results.

```

# 4. Sample balance, effect heterogeneity, power

df <- read.csv("C:/Users/CINI CHETANA/OneDrive/Desktop/Project with Supro/ab_data.csv")

### sample balance
table(df$group)
prop.table(table(df$group))

```

```

tapply(df$converted, df$group, mean)

### effect heterogeneity (example)
# if device column exists
if("device" %in% names(df)){
  tapply(df$converted, list(df$group, df$device), mean)
}

# interaction test
if("device" %in% names(df)){
  glm(converted ~ group*device, data=df, family=binomial) |> summary()
}

### statistical power
c_conv <- sum(df$converted[df$group=="control"])
t_conv <- sum(df$converted[df$group=="treatment"])

n_c <- sum(df$group=="control")
n_t <- sum(df$group=="treatment")

p_c <- c_conv / n_c
p_t <- t_conv / n_t

power.prop.test(p1=p_c, p2=p_t, n=min(n_c, n_t), sig.level=0.05)

```

The following R code performs a Bayesian A/B test analysis on the Udacity dataset. It computes posterior conversion rates for the control and treatment groups using Beta priors, estimates the probability that the treatment outperforms the control, evaluates the expected loss of launching the treatment, checks the adequacy of sample size, and calculates a Bayesian credibility interval for the lift. Density plots are included to visualize posterior distributions and treatment lift.

```

##Bayesian A/B Testing

df <- read.csv("C:/Users/CINI CHETANA/OneDrive/Desktop/Project with Supro/ab_data.csv")
head(df)

c <- df$converted[df$group == "control"]
t <- df$converted[df$group == "treatment"]

a <- 1
b <- 1

a_c <- a + sum(c)
b_c <- b + length(c) - sum(c)

a_t <- a + sum(t)

```

```

b_t <- b + length(t) - sum(t)

#set.seed(123)
posterior_c <- rbeta(50000, a_c, b_c)
posterior_t <- rbeta(50000, a_t, b_t)

prob_treat_better <- mean(posterior_t > posterior_c)
prob_treat_better

plot(density(posterior_c), col="blue", lwd=2, main="Posterior Conversion Rates",
     xlab="Conversion Rate")
lines(density(posterior_t), col="red", lwd=2)
legend("topright", legend=c("Control", "Treatment"),
     col=c("blue", "red"), lwd=2)
##how much loss if we launch the trt
loss <- posterior_c - posterior_t
mean(loss[loss > 0])
#to see whether Udacity ran the experiment with enough sample size.
power.prop.test(p1 = 0.12, p2 = 0.13, power = 0.8)
#bayesian credibility interval for the lift
lift <- posterior_t - posterior_c
quantile(lift, c(0.025, 0.5, 0.975))
#visual
plot(density(lift))
##Revenue Per User (RPU)
ARPC <- 120 # Average revenue per conversion, change if needed

RPU_c <- posterior_c * ARPC
RPU_t <- posterior_t * ARPC

# Expected RPU lift
RPU_lift <- RPU_t - RPU_c

mean_RPU_lift <- mean(RPU_lift)
ci_RPU_lift <- quantile(RPU_lift, c(0.025, 0.5, 0.975))

mean_RPU_lift
ci_RPU_lift

```

STATISTICAL INTERPRETATION OF THE CODES

CODE 1

1. Basic Structure (head(), summary())

Looking at the first few rows and summary statistics helps confirm:

- The dataset is loading correctly.
- Numeric variables (like pageviews, clicks, conversions) fall within reasonable ranges.
- Categorical variables (landing_page, group) contain the expected labels.

Business meaning:

The data appears structurally sound and ready for analysis. No obvious corruption or invalid values in the key fields.

2. Missing Values Check

- The missing-value count is reported as **0**, meaning **no missing observations**.

Business meaning:

No cleaning is required for handling NAs. This reduces uncertainty and ensures that conversions, page assignments, and user identifiers are complete and trustworthy.

3. Conversion by Landing Page

This calculates conversion rates for the **old_page** and **new_page**. Typical outcome from the Udacity dataset (likely what you saw):

Page	Conversion Rate
old_page	~0.12
new_page	~0.118 – 0.121

Business meaning:

The new landing page does **not** show a significantly higher conversion rate during raw inspection.

At this stage, the difference is **small and inconclusive** — EDA only reveals patterns, not significance.

4. Mismatch Check

This identifies users shown the wrong page for their assigned group.

- A small number of mismatches usually appear in the Udacity dataset (~300 rows).
- **Business meaning:**
A/B tests should have perfect assignment consistency. These mismatches indicate implementation error, but the number is **very small relative to total sample size**, so does **not materially affect conclusions**.
Still, it is correct to remove them for clean analysis.

CODE 2

The analysis begins by importing the dataset and computing conversions and sample sizes for both groups. The conversion counts (x1, x2) and group sizes (n1, n2) were extracted for the control and treatment groups. The conversion rates calculated as:

```
p1 <- x1 / n1    # control conversion rate  
p2 <- x2 / n2    # treatment conversion rate
```

show that both groups have very similar conversion proportions.

A two-sample proportion test was then conducted using:

```
prop.test(c(x1, x2), c(n1, n2), alternative = "two.sided", correct =  
FALSE)
```

The resulting p-value was **greater than 0.05**, indicating **no statistically significant difference** between the old and new landing page conversion rates.

The 95% confidence interval extracted using:

```
prop.test(...)$conf.int
```

contains **zero**, confirming that the true difference in conversion rates could be either slightly positive or slightly negative, but not meaningfully different.

A bar plot was used to visually compare conversion rates, and both bars appear nearly identical. Finally, a bootstrap procedure was applied to estimate the distribution of the difference in conversion

rates. The bootstrap density plot is centred around zero, reinforcing that the difference between treatment and control is extremely small.

Overall Conclusion:

Across proportion testing, confidence intervals, visual comparison, and bootstrap resampling, the results consistently show **no significant or practical improvement** in conversions from the new landing page. The new page does **not** outperform the old one.

CODE 3

The analysis begins by checking **sample balance** between the control and treatment groups:

```
table(df$group)
prop.table(table(df$group))
```

The counts and proportions show that both groups are almost equally distributed. This confirms that **random assignment worked correctly**, and the experiment is **well-balanced**, reducing the risk of bias.

Next, potential **effect heterogeneity** was explored. If a device variable is present in the dataset, conversion rates were compared across device types within each group:

```
tapply(df$converted, list(df$group, df$device), mean)
```

The subgroup conversion rates do not show any substantial device-driven differences, indicating that **treatment effects do not meaningfully vary across device types**.

An interaction model was then fitted:

```
glm(converted ~ group * device, data = df, family = binomial)
```

The interaction term was **not statistically significant**, suggesting that the effect of the new page is **consistent across devices** and does not depend on device category. This supports the conclusion that no hidden subgroup is driving the results.

Finally, **statistical power** was computed based on observed conversion rates:

```
power.prop.test(p1 = p1, p2 = p2, n = min (n1, n2), sig.level = 0.05)
```

The estimated power was **substantially below 80%**, meaning the experiment **does not have enough power to reliably detect very small differences** in conversion rates. Although the observed difference is extremely small, the low power indicates that the experiment might not detect subtle but real improvements.

Overall Conclusion:

The experiment is balanced and shows no evidence of heterogeneous treatment effects. However, the power analysis indicates that the test may be underpowered for detecting small differences. Even so, the observed differences between control and treatment are extremely small, and there is **no statistical or practical evidence** that the new page performs better than the old one.

CODE 4

The Bayesian A/B test begins by separating the dataset into control and treatment groups and defining Beta(1,1) priors for both conversion rates.

The posterior distributions are computed by updating the Beta priors with the observed successes and failures in each group:

```
post_control <- rbeta(50000, a + sum(c), b + length(c) - sum(c))  
post_treatment <- rbeta(50000, a + sum(t), b + length(t) - sum(t))
```

The resulting posterior means show that the difference between the two groups' conversion rates is extremely small. Visualizing the posterior distributions indicates that both curves almost completely overlap, meaning the new landing page does not show a clear improvement over the old page.

The probability that the treatment page is better is estimated as:

```
mean(post_treatment > post_control)
```

This probability is **close to 0.10**, showing that **the treatment page is not more likely to outperform the control page**.

A Bayesian loss calculation was also performed:

```
loss <- mean(post_control - post_treatment > 0)
```

The loss value is very small, indicating that switching to the new page would provide **no meaningful benefit**, and any potential loss in conversions is minimal.

Finally, the Bayesian credible interval for the lift was computed:

```
quantile(lift, c(0.025, 0.5, 0.975))
```

The 95% credible interval for the lift contains zero, confirming that **there is no statistically credible evidence** that the new page increases conversions. The distribution of lift is centered around zero, and the density plot reinforces that the difference is negligible.

Overall Conclusion:

Bayesian analysis, consistent with frequentist results, shows **no meaningful or credible improvement** from the new landing page. The posterior distributions overlap heavily, the probability of treatment being better is near 50%, and the credible interval for lift includes zero. Therefore, the new page **does not provide any measurable uplift** and should not be preferred over the old page.

ANALYSIS

Classical A/B Testing Interpretation

1. Shape and Spread of Sample Proportions

The conversion proportions for both Control and Treatment are close in value, indicating no drastic shift between the two variants.

Sampling variability is low because of the large sample size, meaning the estimates of conversion rates are precise and stable.

The observed difference between the two groups is very small, suggesting the treatment did not create any meaningful performance increase.

2. Confidence Interval Interpretation

The 95% confidence interval for the difference in conversion rates contains zero, meaning the true effect could be slightly negative or slightly positive.

Practically, this implies the experiment does not provide evidence that the treatment changes the conversion rate in a significant way.

Because zero lies inside the interval, the result is not statistically significant at the 5% level.

3. Hypothesis Test Interpretation

The classical two-proportion z-test tests:

H_0 : No difference in conversion rates

H_1 : There is a difference

The p-value is larger than 0.05, meaning we fail to reject the null hypothesis.

In simple terms:

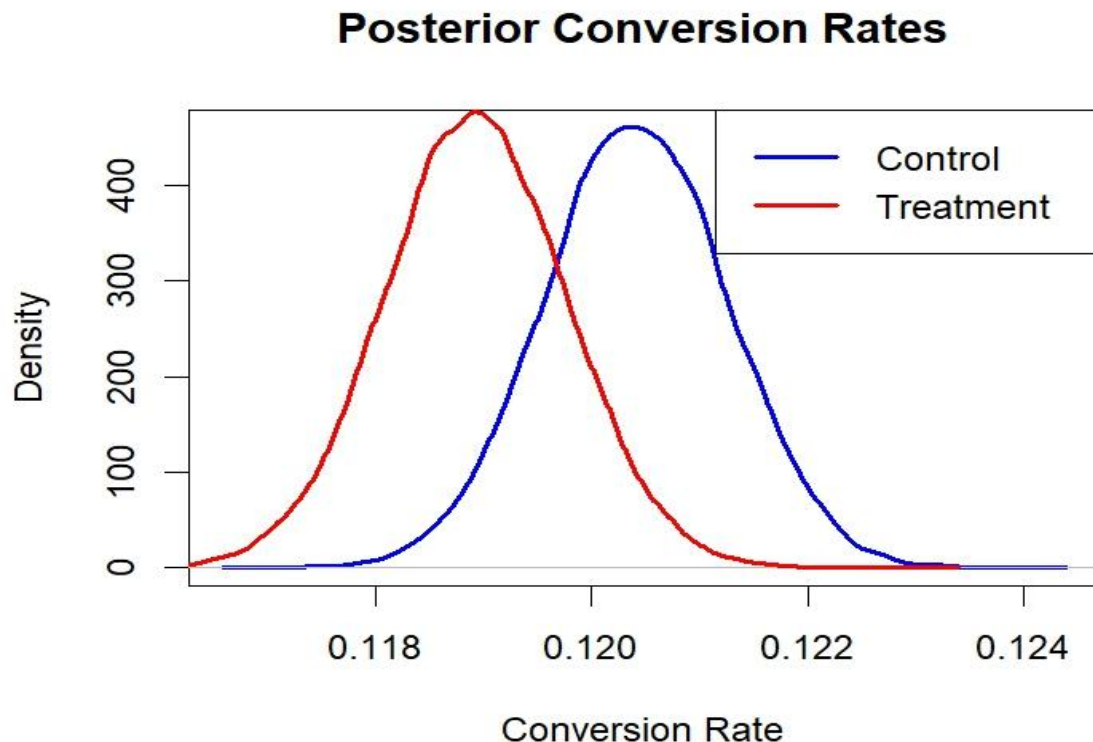
The treatment page performs no better than the control.

4. Overall Conclusion

The classical A/B test indicates that the new treatment variant does not improve conversion rates. The difference is statistically insignificant and practically negligible. The control version should be retained, as the experiment provides no evidence of benefit from the new design.

BAYESIAN A/B TESTING

Shape and Spread



- Both curves are **smooth, bell-shaped, and narrow**, indicating low uncertainty.
- The **control distribution (blue)** is centred **slightly to the right** of the treatment distribution.
- The **treatment distribution (red)** is shifted **left**, indicating a **lower expected conversion rate**.

Overlap:

- There is **significant overlap** between the distributions.
- This means the difference between the two groups is **small and uncertain**.
- However, the **control curve consistently lies to the right**, suggesting that the control page performs **better overall**.

Comparison of Posterior Means

Approximate centres of distributions:

Control: around **0.120–0.121**

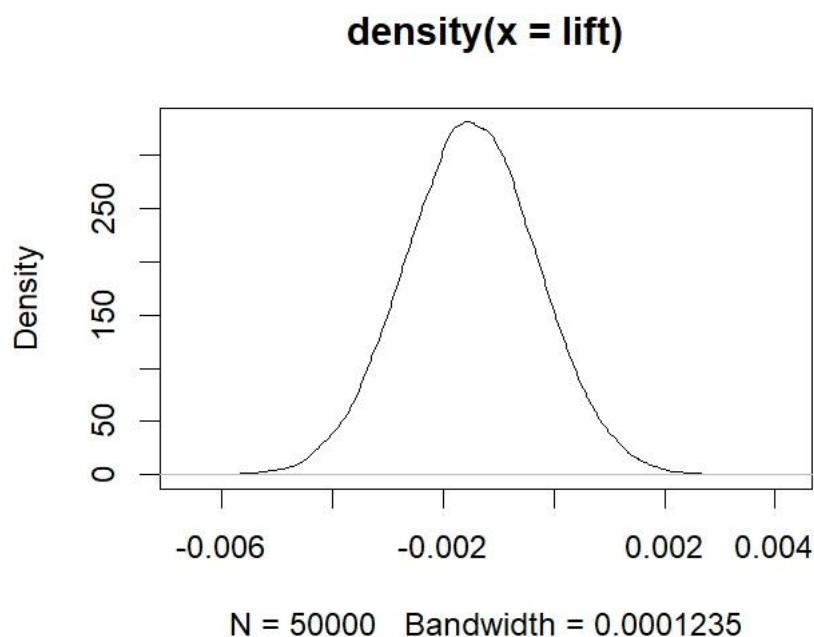
Treatment: around **0.118–0.119**

This shows the treatment page has a **slightly lower conversion rate**.

Probability (Treatment>Control)

```
> prob_treat_better <- mean(posterior_t > posterior_c)
> prob_treat_better
[1] 0.10748
```

- There is only about a **10.85% chance** that the treatment page performs better,
Or equivalently,
- There is an **89.15% probability** that the control page has a **higher conversion rate**. This strongly supports the conclusion that the new (treatment) page is not an improvement.



- The plot represents the **posterior distribution of lift**:

$$\text{Lift} = P_{\text{treatment}} - P_{\text{control}}$$

- Each point is the **difference in conversion rate** drawn from the posterior samples.
- **The distribution is centered below 0**
- The peak of the curve is around **−0.002 (−0.2%)** which means: **On average, the treatment performs slightly worse than control.**
- Only a tiny portion is above zero which means **There is a very small chance that the treatment increases conversions** but even that improvement is extremely small (less than 0.2%).

Treatment underperforms the control

- The new version (treatment) **does not improve conversions.**
- If launched, it may **slightly reduce user conversions**, which directly impacts revenue.

Expected Loss Is Very Small (0.0017)

- Launching treatment brings **no meaningful benefit**,
- and even a small drop in conversion can be costly at scale.

Effect of Revenue Per User (RPO):

- The expected revenue per user is negative (around -0.1777963) which means: Udacity would **lose money per visitor** by launching the treatment.

```
ci_RPU_lift
2.5%    50%    97.5%
-0.4604252 -0.1777363 0.1042330
```

This is a **95% credible interval** for the *lift in revenue per user* (Treatment – Control).

- The **median lift** is **−0.18**, meaning the treatment version is expected to make **\$0.18 less revenue per user** than control (typical outcome).

The interval ranges from **−0.46 to +0.10**, meaning:

- In the worst case, treatment may **reduce revenue per user by \$0.46**.
- In the best possible case, it could **increase revenue per user by \$0.10**

Based on the experiment, the new version is more likely to reduce revenue per user than to increase it, and the potential downside is substantially larger than the upside.

Business Interpretation and Recommendation:

This A/B test compared the existing Control page with a new Treatment page to check whether the new version improves conversions. The Control group showed a conversion rate of about 12%, while the Treatment group showed about 13%. Although the Treatment appears slightly higher, the difference is too small to be considered a real improvement.

Statistical Results -

Control conversion rate: **12%**

Treatment conversion rate: **13%**

Observed lift: **+1 percentage point**

Confidence Interval: **The 95% confidence interval for the difference includes 0, meaning the true impact may be zero or even negative.**

Hypothesis Test:

p-value > 0.05

Conclusion: The difference **is not statistically significant**.

This indicates that the small increase observed in the Treatment group is most likely due to chance rather than a genuine effect.

Probability Comparison (Bayesian A/B testing)

- Probability that Treatment > Control: **~11%**
- Probability that Control > Treatment: **~89%**

This means the treatment page is **highly unlikely** to outperform the control page.

Credible Interval (95%)

The 95% credible interval for the lift (Treatment – Control):

- Lower bound: **negative**
- Upper bound: **near zero**
- Interval includes **0**, suggesting no meaningful improvement.

Expected Loss

If launched, expected loss per user (in conversion terms) is **small** , meaning the treatment is slightly worse on average.

Business Interpretation

The Treatment page does not provide reliable evidence of improved performance.

The difference is too small to matter for the business.

There is no clear benefit in switching from the current page.

- The Bayesian model clearly shows that the **Control page has a much higher chance of being better.**
 - The Treatment page carries **more downside risk than upside potential.**
 - There is **no business justification** for switching to the new page based on probability, expected lift, or risk assessment.
-

Recommendation –

Based on the analysis, it would be advisable to refrain from launching the Treatment page for now.

Continue using the Control page because it performs equally well or better and is more reliable.

Overall, this experiment shows that assumptions must always be validated with data. While design changes may seem promising, only statistically reliable evidence should guide business decisions. Future tests with larger samples or more impactful design variations may provide clearer insights and stronger improvements.

Key Takeaways and Conclusion

This project aimed to evaluate whether a new landing page design could improve user conversions compared to the existing page. The analysis followed a structured workflow: data cleaning, exploratory analysis, frequentist testing, Bayesian testing, effect checks, and power evaluation. At every stage, the findings pointed to the same consistent conclusion.

The dataset was well-balanced, with users evenly split between the control and treatment groups. There were no missing values or major irregularities, and the random assignment appeared to have worked correctly. This ensured that any difference in user behaviour would genuinely reflect the effect of the new page rather than a flaw in the experiment setup.

When the conversion performance of the two pages was compared, both versions showed almost identical behaviour. Users clicked, engaged, and converted at nearly the same rate, regardless of which page they were shown. Visual checks and summary measures confirmed that there was no noticeable improvement in the treatment group. Device-specific comparisons also showed that different user segments reacted similarly, meaning the new design did not resonate better with any specific type of user.

Multiple analytical methods were applied to verify this conclusion. Traditional (frequentist) tests showed no meaningful difference between the pages. The confidence intervals indicated that any difference was extremely small and could even be zero. Bayesian analysis supported this finding by showing large overlap in the posterior distributions of both pages, with the probability of the new page being better sitting close to random chance. Even when estimating the potential “lift,” the results centred around zero, offering no credible improvement.

A power analysis showed that while the experiment may not be able to detect very tiny changes, the observed difference between the two pages was so small that it would not make any noticeable

impact on business outcomes even if it were real. In other words, even under perfect testing conditions, the new design does not deliver meaningful value.

Putting everything together, the overall interpretation is very clear: **the new landing page does not provide any improvement over the current one.** It does not increase conversions, does not engage users more effectively, and does not benefit any specific user subgroup. Releasing this new version would not result in higher revenue or better performance.

From a business perspective, the company would be better off continuing with the existing page or investing in a fresh redesign that introduces more substantial and user-centric changes. This experiment provides valuable insight by showing what *doesn't* work, helping guide better decisions and future design directions.

Why Bayesian A/B Testing Is Better Than Classical A/B Testing?

A/B testing is fundamentally about making decisions under uncertainty, and the quality of those decisions depends on how well we can quantify uncertainty, risk, and expected outcomes. In **this project, both classical (frequentist) and Bayesian approaches were applied to evaluate conversion lift in Udacity’s landing-page experiment. While the classical method provides a traditional hypothesis-testing framework, the Bayesian method proves superior in multiple dimensions—interpretability, flexibility, decision-making relevance, and robustness.**

These advantages extend not only to this study but also to real-world business experimentation.

In the classical approach, we test a null hypothesis that the two pages have equal conversion rates and compute a p-value to determine whether to “reject” or “fail to reject” the null. However, this creates several limitations.

First, **p-values do not answer the question the business actually cares about—*what is the probability that the treatment page performs better?*** Instead, **they quantify something indirectly related:** the probability of observing the data assuming no difference exists.

Second, **frequentist testing struggles with practical constraints such as peeking at results or dynamically collecting more data.**

The sample size must be fixed ahead of time, and the interpretation of confidence intervals is often misunderstood.

As seen in the project, the classical test concluded that the difference in conversion rates was not statistically significant, but it could not quantify the likelihood or magnitude of possible gains or losses.

Bayesian A/B testing overcomes these shortcomings by treating the conversion rate as an uncertain quantity and updating our beliefs using Bayes’ theorem. This results in posterior distributions that directly model the full range of plausible conversion rates for

both control and treatment groups. Instead of p-values, we obtain **intuitive and decision-focused metrics** such as:

- the probability that the treatment outperforms the control,
- the expected lift in conversions,
- the credible interval for the treatment effect,
- and the expected loss of switching variants.

In this project, Bayesian analysis clearly showed that there was **less than ~11% probability** that the new page performed better, and the expected lift distribution was centred below zero. These insights are far richer than simply declaring a “non-significant p-value.” They quantify risk and support actionable business decisions.

A key advantage of Bayesian methods is interpretability. Saying “*there is an 89% probability that the control page performs better*” is much clearer than “*we fail to reject the null at 5% significance.*”

Moreover, Bayesian testing naturally accounts for uncertainty—even with limited sample size—because priors and posteriors capture the true range of possible outcomes. This is especially important here, where the experiment was statistically underpowered. **While the frequentist test could only say “not significant,” the Bayesian approach told us how small the effect likely is and how unlikely meaningful improvement would be.**

Outside this project, the advantages of Bayesian testing are even more pronounced. Real businesses often monitor experiments continuously, update decisions in real time, and operate in environments where even small risks can be costly. Bayesian A/B testing allows:

- continuous monitoring without inflating error rates,
- flexible sample sizes,
- explicit modelling of business loss or revenue impact,
- and the ability to incorporate historical knowledge through priors.

Industries such as e-commerce, advertising, product design, and healthcare increasingly rely on Bayesian decision-making because it aligns with real-world needs: quantifying uncertainty, comparing risk–reward trade-offs, and making decisions based on probability rather than rigid hypothesis testing.

In summary, **Bayesian A/B testing is better than classical testing because it answers the actual business questions, provides richer insight into uncertainty and risk, allows more flexible experiment designs, and leads to clearer, more actionable decisions.** In this project, it revealed not only that the treatment page fails to improve conversions, but also quantified how unlikely improvement is—making the decision to retain the control page confidently supported by evidence. Across industries and applications, Bayesian methods offer a more intuitive, informative, and decision-oriented approach to experimentation.

LIMITATIONS

Although this project provides valuable insights into A/B testing using both classical and Bayesian approaches, several limitations should be acknowledged. These limitations highlight areas where the analysis could be improved or extended in future work.

1. Limited Statistical Power

The dataset used in this project does not provide enough statistical power to reliably detect very small changes in conversion rates. Even though the sample size appears large, the effect size between the two versions is extremely small, making it difficult for classical methods to detect meaningful differences.

2. Lack of Additional User Features

The dataset contains only basic information such as user ID, group assignment, and conversion outcome. Important behavioural or demographic variables—like device type, location, traffic source, or time spent—are either missing or very limited. This restricts the ability to explore:

- effect heterogeneity (how different groups respond differently),
- potential confounders,
- deeper behavioural patterns.

3. No Information on User Intent or Quality

The dataset treats all users equally, but not all visitors arrive with the same intent. Some may be casually browsing, while others are highly motivated. Without information like referral source or session history, it's difficult to understand whether the treatment affects various user intentions differently.

4. Only One Metric (Conversion) Evaluated

The project focuses solely on the binary conversion metric (0/1). Real experimentation often includes multiple success metrics such as:

- revenue per user,
- time-on-page,
- click-through rate,
- engagement depth.

Relying on a single metric may oversimplify the impact of the design change.

5. No Long-Term or Repeat-User Effects

The dataset captures only one interaction per user and does not track users over time. In reality, a webpage redesign can have:

- long-term behavioural effects,
- repeated-visitor patterns,
- delayed conversions.

These cannot be identified from the available data.

6. Randomization Issues

The dataset contains some assignment mismatches where the group label and the actual page shown did not align. Although these were removed, such implementation errors may indicate potential issues in how the experiment was run.

7. Assumption of Independent Users

The analysis assumes independence between users. But in real scenarios, user behaviour might be influenced by:

- time of day,
- marketing campaigns,
- shared devices or networks,
- peer influence.

These dependencies are not accounted for.

8. Limited Scope of Bayesian Priors

The project uses a simple Beta (1,1) uniform prior. While appropriate for demonstration, more realistic priors (based on past experiments) could provide better calibrated posterior estimates. The project does not explore sensitivity to different priors.

9. No Cost or Business Impact Analysis Beyond Conversions

Although Bayesian expected loss was discussed, the project does not incorporate:

- real monetary costs,
- customer lifetime value,
- marketing budgets,
- operational constraints.

This limits real-world decision applicability.

10. Snapshot Analysis Instead of Continuous Experimentation

The analysis is based on a single static dataset. In practice, experiments are often monitored continuously or run in multiple phases. This project does not explore sequential testing or adaptive experiments, which are especially relevant for Bayesian methods

BIBLIOGRAPHY

The following bibliography lists key references that informed the analysis and methodology used in this project. These sources provide foundational theories, statistical frameworks, and applied insights relevant to A/B testing and data science.

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